

The empirical recurrence for OEIS A277941, recovered from its tail

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8 September 2026

Abstract

OEIS A277941 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 126$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{70} = 595289941475328 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A277941 is “Number of $n \times 4$ 0..2 arrays with every element plus 1 mod 3 equal to some element at offset (-1,0) (-1,1) (0,-1) (0,1) or (1,0), with upper left element zero.”. It has offset 1 and begins

0, 74, 2353, 84970, 2967962, 103960169, 3638916907, 127378891305, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Nov 05 2016 11:36:43, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 126$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 126$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 252$ exact terms from $n = 4$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	53	c_{19}	−140453296858882	c_{37}	−291824074577380411204	c_{55}	1537201573995215
c_2	−680	c_{20}	−386610066778231	c_{38}	454204905003484655161	c_{56}	3926190094906007
c_3	1018	c_{21}	1416874506742538	c_{39}	1377652086142368570367	c_{57}	2015891696747064
c_4	28770	c_{22}	5407336469578164	c_{40}	−964940627171050367291	c_{58}	1270273451179434
c_5	−88152	c_{23}	−10937584463879768	c_{41}	−1888572232564371578809	c_{59}	−127314164045321
c_6	−707443	c_{24}	−58066839511452853	c_{42}	1201277011375461476493	c_{60}	−841691590051641
c_7	1223658	c_{25}	63866754074270077	c_{43}	200349807758676927072	c_{61}	2425966592675674
c_8	17739875	c_{26}	508321915094705677	c_{44}	−46401543739933204459	c_{62}	1252561483521288
c_9	22015513	c_{27}	−327935617113493103	c_{45}	2032394683850985554265	c_{63}	−75492550779393
c_{10}	−415947627	c_{28}	−3455430211991361951	c_{46}	−630808410284146818852	c_{64}	43149946413268
c_{11}	−1470986943	c_{29}	1417143422239815122	c_{47}	−2429951681079816053890	c_{65}	241569756169109
c_{12}	7833761976	c_{30}	18300234071817217254	c_{48}	194223779072500858514	c_{66}	58870851466649
c_{13}	37349318410	c_{31}	−4833567867150544247	c_{49}	218373065817845673164	c_{67}	−25735258065056
c_{14}	−85062469942	c_{32}	−75137350141603958750	c_{50}	−7207336488803723000	c_{68}	−1364744788888
c_{15}	−704083492400	c_{33}	15059870276893961628	c_{51}	1576724255223340133842	c_{69}	78544215075717
c_{16}	29676260168	c_{34}	216406647969114543681	c_{52}	448927476466543145872	c_{70}	5952899414753
c_{17}	11058703677317	c_{35}	−11833916243536876802	c_{53}	−1115966920354784629668		
c_{18}	17327582693758	c_{36}	−372699044211224503035	c_{54}	−400721912383445743952		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A277941 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = -595289941475328$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 595289941475328.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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